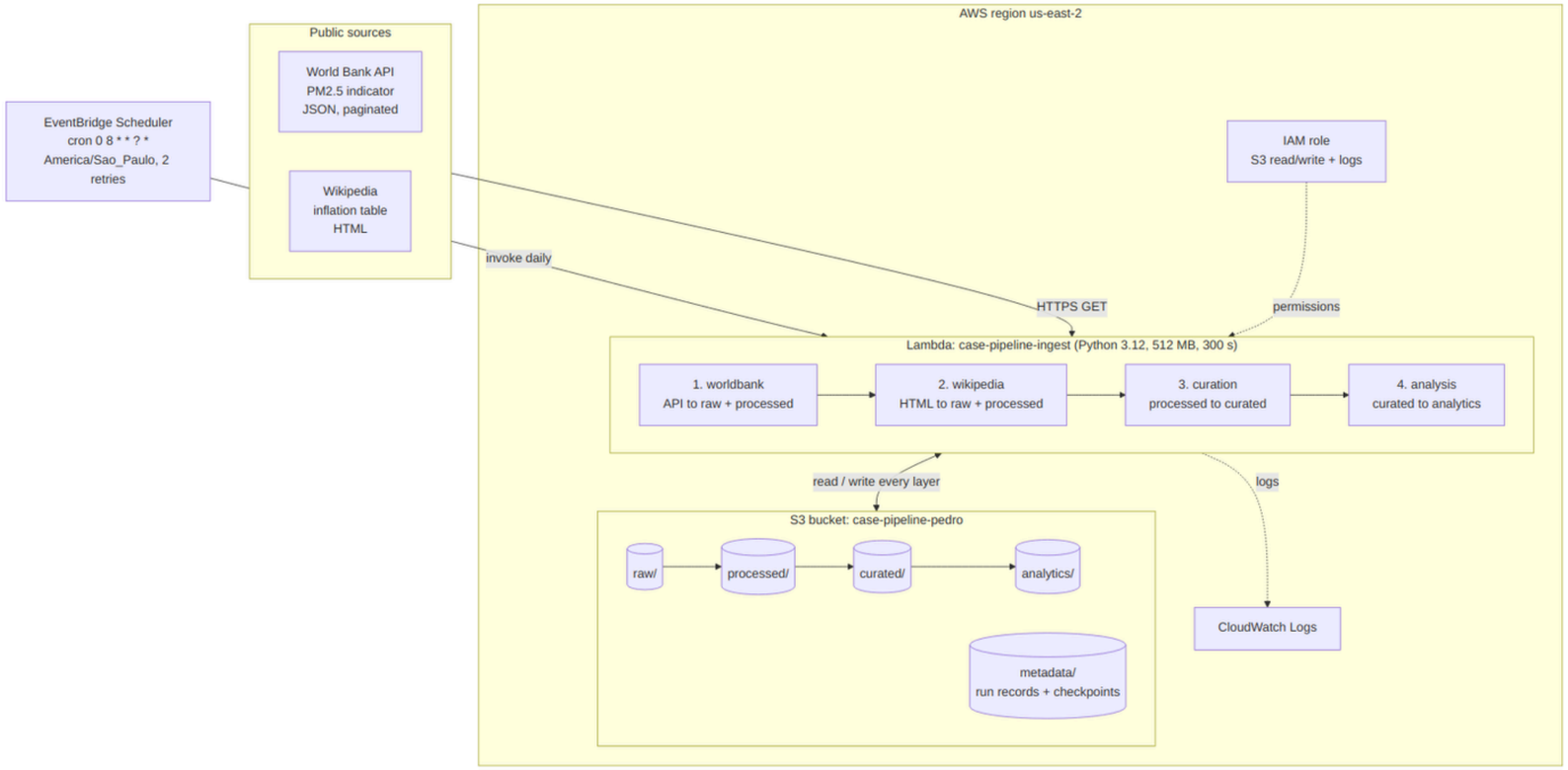
Air Pollution vs. Inflation

A cloud-native, incremental data pipeline on AWS
World Bank API + Wikipedia crawler | Python, S3, Lambda, EventBridge

Architecture: serverless and event-driven



- EventBridge triggers one Lambda every day
- Four ordered stages: World Bank, Wikipedia, curation, analysis
- S3 layers: raw, processed, curated, analytics (+ metadata)
- ~116 MB and ~3 s per run when sources are unchanged
- Least-privilege IAM role; no keys in code

Incremental ingestion: only process what changed

World Bank API

- Compares the source's lastupdated with the checkpoint
- Fetches only the needed year window (+2-year refresh)
- Paginated (1,000 rows/page), upsert by country + year

Wikipedia crawler

- Hash of the raw table content vs. checkpoint
- Edits elsewhere on the page do not trigger a reload
- Upsert by country + year

Curation

- Fingerprint of the three processed inputs
- Join rebuilt only when inputs changed
- Unmatched countries reported, never silently dropped

- Idempotent: a re-run on unchanged data inserts 0 rows and keeps the rest unchanged
- Checkpoints advance only after a stage succeeds; failures are recorded and retried next run
- Every run writes a metadata record: timestamp, source, row counts, duration, status

Insight: a weak positive association

+0.28

Spearman correlation

+0.08

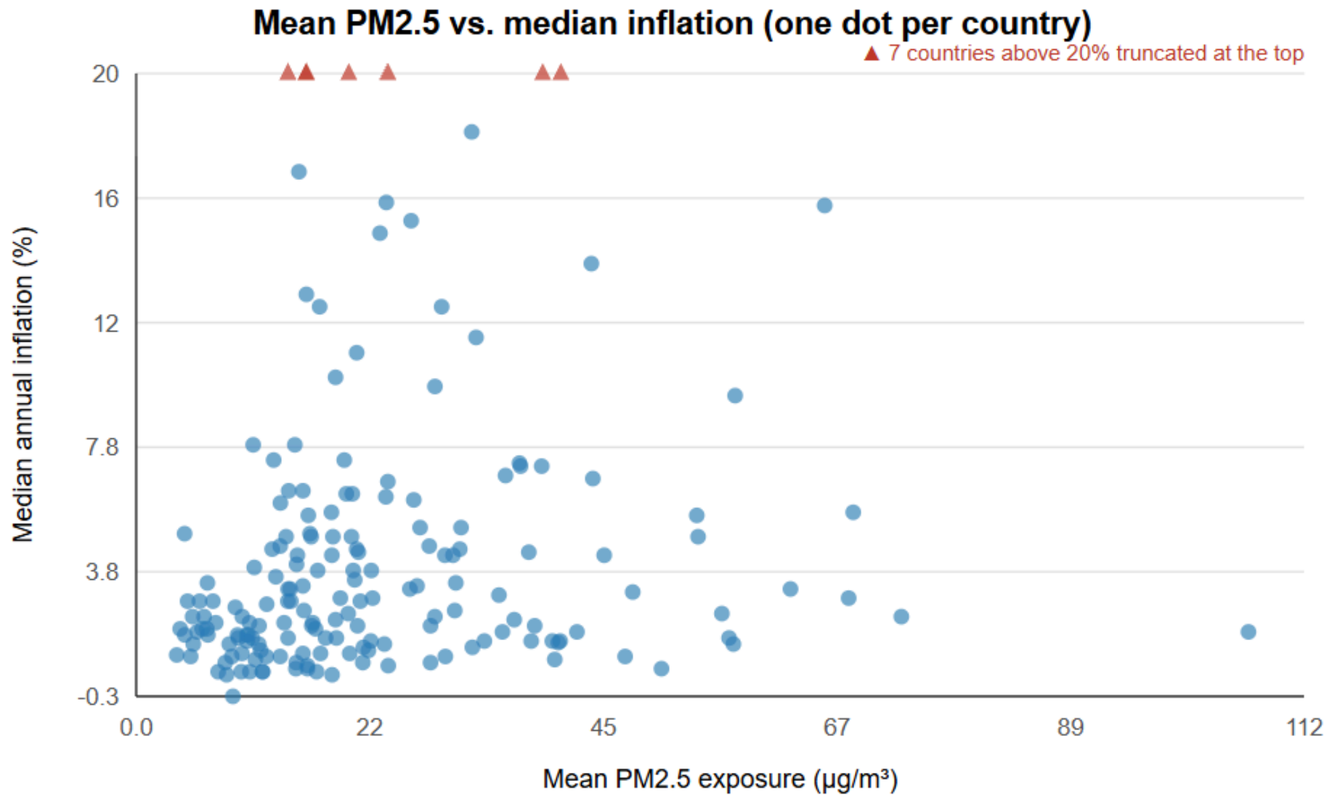
Pearson correlation

178

countries compared

1,583

country-year rows, 2015-2023



- Countries with higher mean PM2.5 tend to have somewhat higher median inflation
- Spearman is far above Pearson: a few hyperinflation countries distort the linear fit, while ranks resist them
- An association, not causation: both indicators track country development

How the insight is produced from the curated dataset

1 Curated dataset

country x year table in S3 (curated/)

2 Aggregate

per country: mean PM2.5 and median inflation (min. 3 years)

3 Statistics

Spearman (primary), Pearson, yearly trend, income and region summaries

4 Render

SVG charts, report.html and summary.json in analytics/

Why these choices

- Median and Spearman: inflation has extreme outliers
- Country-level view: one point per country, comparable across years

Why pure Python

- Standard library only (no pandas or matplotlib)
- Same code runs locally and in Lambda; regenerated on every run

Key technical decisions and trade-offs

Decision	Why	Trade-off
Lambda + S3 + EventBridge	Seconds-long workload; no idle cost, no servers	Incremental state is kept in code, not by an orchestrator
Checkpoints as JSON in S3	Simple, inspectable, and the same code runs locally	No cross-run locking (one daily run)
CSV instead of Parquet	Tiny package, no pandas/pyarrow layer, human-readable	Types re-applied on read; not columnar
ISO3 as canonical key + alias dictionary	Names differ between sources; codes are stable	New naming variants need an alias (reported)
Pure-Python SVG/HTML report	No heavy dependencies; works identically in the cloud	Less styling flexibility than a plotting library

Challenges faced and how they were solved

Challenge	Solution
Different country identifiers (names vs. ISO3)	Alias dictionary + normalized names + unmatched report: 189 of 189 matched
Messy HTML table (empty cells, footnotes, unicode minus, -0.0)	Cleaning rules with counters in the run metrics; unit tests with a synthetic fixture
API mixes aggregates with countries and reports errors as HTTP 200	Filter by region (784 rows removed); explicit error detection; retries with backoff
Hyperinflation outliers hide the pattern	Median per country, Spearman as the main metric, truncated chart axis (marked)
Packaging for Lambda from Windows	Build script downloads Linux wheels, so the package runs in the cloud
Staying within a free account	Free Plan, zero-spend budget, tiny footprint, least-privilege IAM